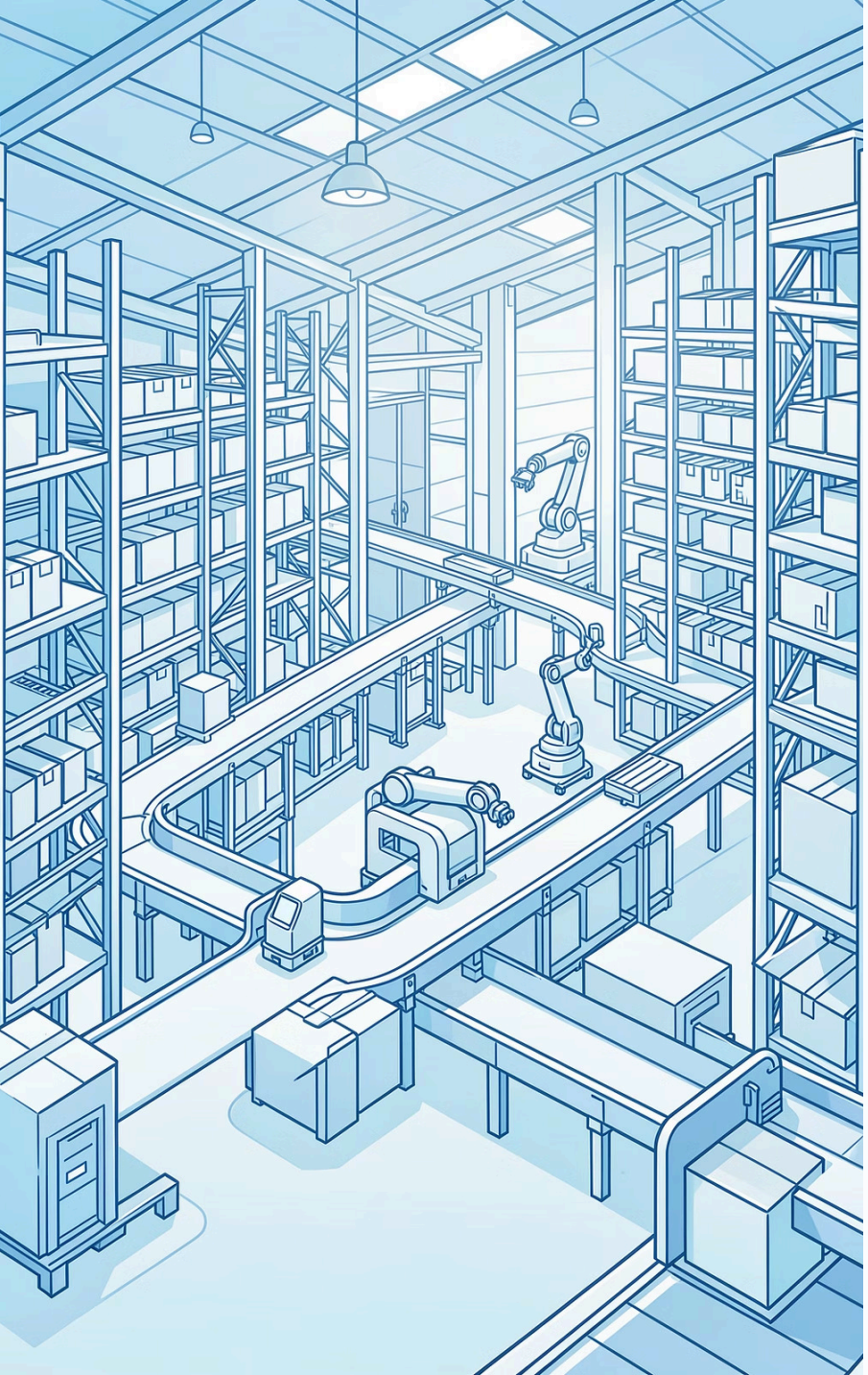

Autonomous Logistics Research Agent using CrewAI & RAG

Submitted By:

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Domain: Agentic AI

Technologies: Python, FastAPI, CrewAI, LangChain, ChromaDB, LLM

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Problem Statement

Warehouses generate large operational data such as:

- Orders
- Delays
- Picking time
- Transport modes

Current issues:

- Manual analysis required
- No instant insights
- Difficult to identify delay causes
- Depends on human analysts

👉 Need an AI system that can analyze and explain logistics operations automatically.

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Objective

To build an **AI-powered autonomous researcher agent** that can:

1. Understand natural language questions
2. Analyze warehouse dataset
3. Perform industry research
4. Provide recommendations
5. Store knowledge for future queries

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Proposed Solution

We developed a **multi-agent AI system** that works like a company team:

Manager Agent

decides task

Data Analyst Agent

analyzes dataset

Research Agent

searches industry knowledge

Memory System

stores learning

The system answers both **data questions and research questions** automatically.

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Technology Stack

Backend

Python, FastAPI

AI Framework

CrewAI, LangChain

LLM

Llama-3 (Groq API)

Vector Database

ChromaDB

Embeddings

Sentence Transformers

Frontend

Streamlit

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System Architecture

Workflow:

01

User asks question

02

Router detects query type

03

Dataset OR Web research agent runs

04

LLM generates structured report

05

Knowledge stored in knowledge base

Uses **Retrieval Augmented Generation (RAG)** to avoid hallucination.

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Dataset Analysis Capability

Example Query:

"Details of order 1003"

👉 **Answer generated directly from data.**

System retrieves record from CSV:

- Warehouse: Indore
- Product: Clothing
- Distance: 560 km
- Delay: Yes
- High picking time

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Research Capability

Example Query:

"Latest warehouse automation technologies"

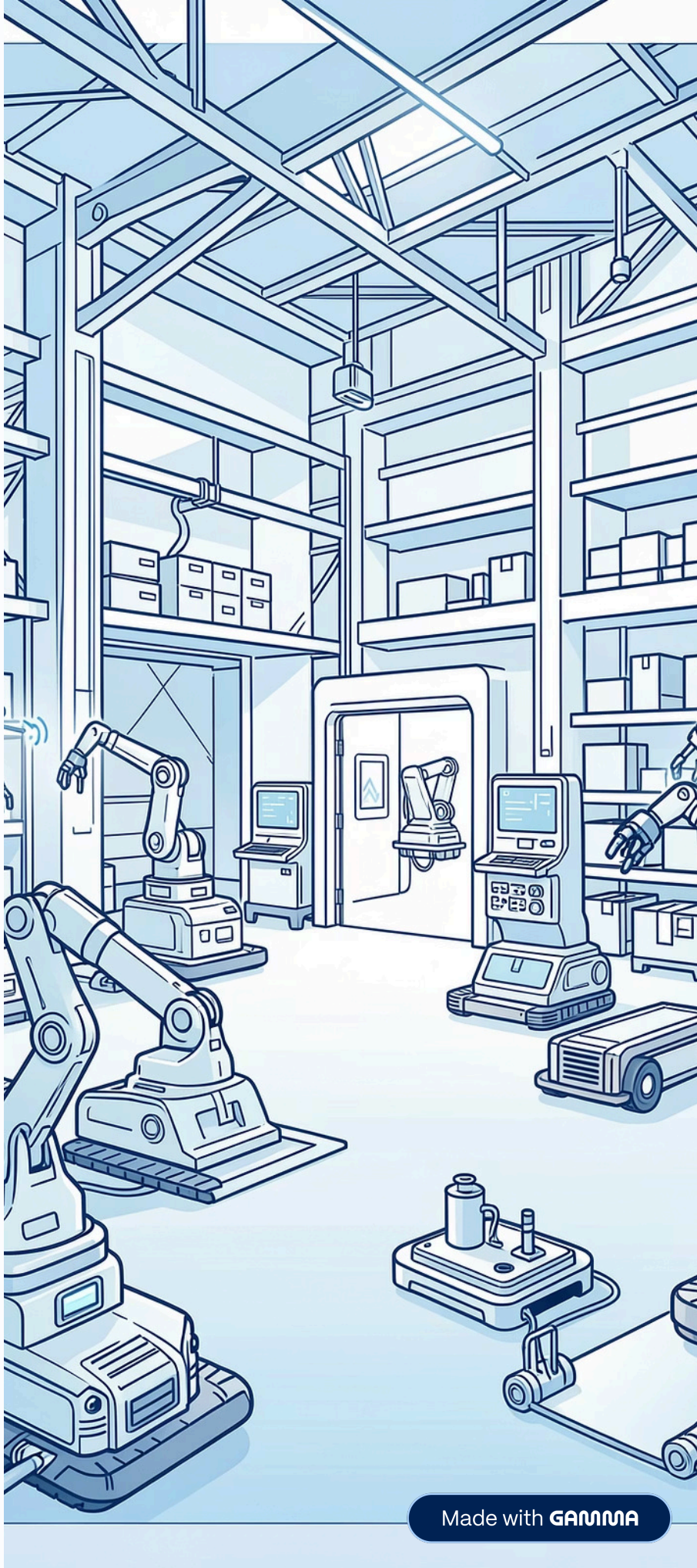
System performs web research and returns:

Robotics picking systems

AI demand forecasting

IoT tracking

Warehouse Management Systems (WMS)



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Output & Advantages

System generates structured report:

- Root Cause
- Insights
- Best Practices
- Recommendations

Advantages

- Automatic analysis
- Faster decisions
- Accurate insights
- Continuous learning

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Conclusion & Future Scope

We built an **Autonomous AI Logistics Analyst** that:

- Understands human queries
- Analyzes warehouse operations
- Researches industry solutions
- Stores knowledge

Future Scope

Real-time IoT integration

Predictive delay alerts

Smart logistics dashboard

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